Purva Rana

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Software engineer with experience building real-time, data-intensive applications across graphics, scientific computing, and AI. Strong foundation in C++, Python, TypeScript, and systems development, with experience taking projects from algorithms and implementation through deployment. MS Computer Science at Duke (Fall 2026).

EDUCATION

Duke University 2026 - 2028

MS Computer Science

MIT World Peace University, Pune 2022 - June 2026

B.Tech. Computer Science and Engineering | CGPA: 8.78 | Merit Scholarship, all four years
Interdisciplinary curriculum blending Computer Science with Peace Studies, Global Cultures, and Ethics.

EXPERIENCE

Research Intern, High Performance Computing - Medical and Bioinformatics Group [HPC-MBA] Feb 2026 - June 2026

Centre for Development of Advanced Computing [C-DAC], Pune

- Computational biologists working with MD simulations typically rely on flat 2D trajectory viewers that flatten structural geometry.
- MolecularVR is a browser-native VR environment where structural biologists and computational chemists can explore conformational changes, binding sites, and molecular interaction dynamics spatially, thereby building the structural intuition needed to inform experimental decisions.
- Wrote the protein cartoon renderer from scratch: Catmull-Rom splines, Rodrigues parallel transport with sign-coherence, DSSP secondary structure integration, Gaussian-smoothed side vectors.
- UI built in vanilla TypeScript for render-loop compatibility: retractable panels, per-layer molecular visibility controls, share and screenshot modal.
- Full deployment: Vercel frontend, Dockerized Node.js/Express + Python (MDAnalysis) backend, SQLite for session persistence.

Tech Advisor, IRIS Club 2024 - 2026

- Led technical direction for the college tech club. Pushed design-first engineering across the team, which brought a 40% lift in platform engagement.

SELECTED PROJECTS

histARy AR Capstone Research

Designed and built an AR reconstruction pipeline for digitally preserving eroding historical structures. Addressed persistent outdoor tracking drift with geometry-driven multi-target tracking, resulting in stable, high-precision overlays on mobile devices.

MindscapeVR VR and Simulation

VR environment for neurosurgical training built in Unity, with 3D neural models authored in Blender. Focused on getting anatomical scale and spatial relationships right so trainees could navigate the structures.

Sustainable Code Gen HPC Research

Fine-tuning LLMs on the CodeNet corpus to generate performant C++, targeting runtime and memory efficiency alongside correctness.

SKILLS

Graphics and Spatial: Three.js, WebXR, WebGL, Unity (C#), Blender
Languages and Systems: TypeScript, Python, C++, Java, Node.js, Docker, Linux, Git
Scientific Computing: MDAnalysis, molecular dynamics workflows, HPC pipelines
AI / ML: PyTorch, TensorFlow - applied work in computer vision and NLP
Spoken Languages: English, Hindi, Marathi, Gujarati (fluent); French and Japanese (in progress)

ACHIEVEMENTS

- Winner, HackMIT Ideathon 2025. Top 5, DataQuest Track at HackMIT Hackathon.
- International Olympiad medals across Informatics, Mathematics, Cyber and Science.
- Highest CS score in cohort (191/200), AY 2021-22. Merit scholarship awarded all four years of undergraduate studies.
- Attended NEPCON Japan 2026, Factory Innovation Week.

RELEVANT COURSEWORK

Data Structures and Algorithms | AI | Machine Learning | NLP | High Performance Computing | AR and VR | Computer Graphics & 3D modelling | Cloud Computing | Computer Networks | Operating Systems | Compiler Design | DBMS | Software Engineering and Project Management